OSANIPIN OLAMITUNBOSUN OLUWANIFEMI

Lagos, Nigeria • nifemibosun70@gmail.com • +234 903 900 5812

github.com/nifemibosun • linkedin.com/in/nifemibosun

PROFESSIONAL SUMMARY

Mathematics student at Lagos State University with strong software development expertise in blockchain, programming language design, and peer-to-peer systems. Proficient in TypeScript, Python, and Rust with hands-on experience building production-grade tools including blockchain networks, Web3 libraries, and distributed systems. Passionate about cryptography, network protocols, and mathematical foundations of computer science.

TECHNICAL SKILLS

Languages: TypeScript (primary), Python, Rust, Go, C

Frontend: React, Vite, TypeScript

Blockchain & Crypto: Bitcoin/blockchain protocols, cryptography libraries, Web3 standards, P2P networking

Tools & Platforms: Git, GitHub, Node.js, Firebase, Supabase, npm

Other Skills: Compiler design, VM architecture, P2P protocols, cryptographic implementation, technical support

WORK EXPERIENCE

Independent Software Developer

Self-Employed • Lagos, Nigeria • 2024 – Present

- Designed and implemented Gyro, a TypeScript-based programming language featuring a complete four-stage compiler pipeline and VM architecture for bytecode execution
- Developed Byte Chain, a blockchain network written in TypeScript implementing Bitcoin-like transaction protocols with P2P networking and cryptographic validation
- Built bc-web3.js, a TypeScript library providing Web3-compatible interfaces for Byte Chain interaction, published on npm for community use
- Created BC Scan, a blockchain explorer frontend using React and Vite, enabling transaction visualization and network inspection for Byte Chain
- Implemented CypherCast, a peer-to-peer messaging system with end-to-end encryption using modern cryptography libraries for secure communication

ICT Intern

Administrative Staff College of Nigeria (ASCON) • Lagos, Nigeria • March 2026 – Present

- Provide technical support for hardware and software issues, diagnosing and resolving system malfunctions for staff and operations
- Troubleshoot hardware components including networking equipment, servers, and peripheral devices
- Support software deployment, system configuration, and user training across the organization

PROJECTS

Gyro Programming Language

TypeScript, Compiler Design, VM Architecture • github.com/nifemibosun/gyro-lang

- A complete programming language implementation featuring lexical analysis, parsing, semantic analysis, and bytecode compilation with a custom virtual machine for execution

ByteChain Test Network

TypeScript, Node.js, Cryptography, P2P Networking • github.com/nifemibosun/ByteChain-Testnet-Alpha

- A blockchain network implementation with Bitcoin-like transaction model, cryptographic signing, peer-to-peer consensus, and network synchronization

ByteChain Wallet

React, Vite, TypeScript • github.com/nifemibosun/bc-wallet

- A frontend application used to create transactions on the ByteChain network.

bc-web3js Library

TypeScript, Web3 Standards, npm • github.com/nifemibosun/bc-web3js

- A TypeScript library providing Web3-compatible methods for interacting with ByteChain, enabling developers to send transactions and query blockchain state

CipherCast - P2P Messaging

TypeScript, P2P Networking, Cryptography • github.com/nifemibosun/cyphercast

- A decentralized peer-to-peer messaging platform with end-to-end encryption, allowing secure communication between participants without central servers

EDUCATION

Bachelor of Science in Mathematics

Lagos State University (LASU) • Lagos, Nigeria • Expected 2025

- Relevant coursework: Abstract Algebra, Number Theory, Graph Theory, Discrete Mathematics, Linear Algebra
- Research interests: Number theory, abstract algebra, cryptographic foundations, mathematical modeling of systems

CERTIFICATIONS

- Hashgraph Developers Course Certificate